

Unit-2

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Bearing Capacity Analysis's.

Rajdhan

DATE / /

Books referred:

1. "Soil Mechanics and Foundation" By Dr. B.C. Punmia.
2. "Foundation Engineering" by Dr. S.B. More, Dr. S.S. Jahagirdar.

Content:

1. Various Terms related to Bearing capacity.
2. Modes of shear failure
3. Methods of determining bearing capacity (Field test for Bearing capacity evaluation)
4. Bearing capacity analysis's.
5. Effect of water table and eccentricity on bearing capacity.
6. Settlement Analysis.

*1. Various terms related to bearing capacity.

Que: Define -

→ 1. Bearing capacity: The supporting power of a soil or rock is referred to as its bearing capacity. OR
It is capacity of the subsoil to bear a load.

2. Ultimate bearing capacity (q_u) -

It is defined as, it is the pressure at the base of the foundation at which the soil or rock fails in shear.

3. Net ultimate bearing capacity (q_{nu}) -

It can be obtained by;

$$q_{nu} = q_u - (\text{Effective surcharge at the base level of foundation})$$

$$\therefore q_{nu} = q_u - \bar{\sigma}$$

$$\therefore \boxed{q_{nu} = q_u - \gamma D}$$

$$\therefore \bar{\sigma} = \gamma D, \quad D = \text{Depth of footing}$$

γ = specific weight of soil.

4. Net safe Bearing capacity (q_{ns}) -
It is the net ultimate bearing capacity divided by a factor of safety (F)

$$q_{ns} = \frac{q_{nu}}{F}$$

5. Safe bearing capacity (q_s) -

$$q_s = q_{ns} + \gamma D \quad \text{OR} \quad q_s = \frac{q_{nu}}{F} + \gamma D$$

6. Permissible settlement Pressure -

It is the load intensity which will cause a permissible settlement for the structure.

7. Allowable bearing capacity - It is the load intensity at which neither the soil fails in shear nor there is excessive settlement.

In other words,

It is the minimum load intensity between net safe Bearing capacity (q_{ns}) and permissible settlement pressure.

* 2. Modes of shear failure

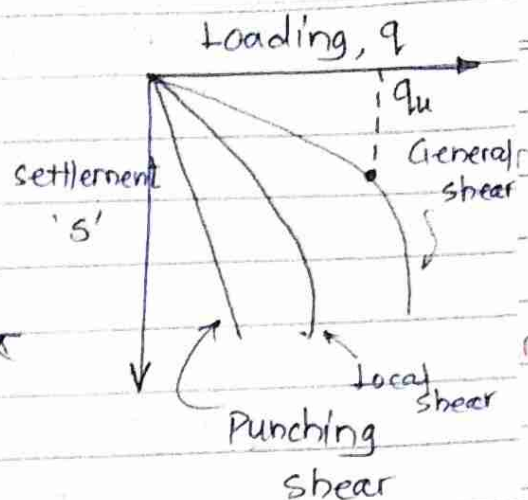
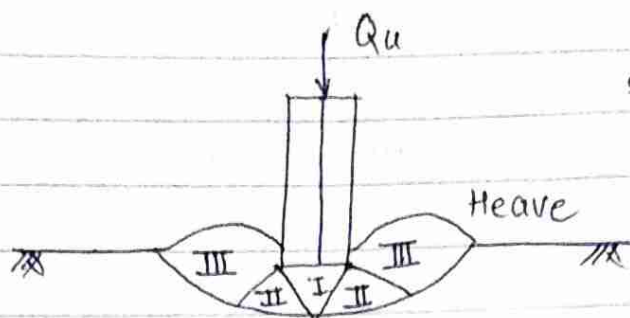
Que: Explain with neat sketches modes of shear failure in soil.

→ Shear failure of soil under Foundation base can take place in three different ways depending up on type of soil and location of foundation.

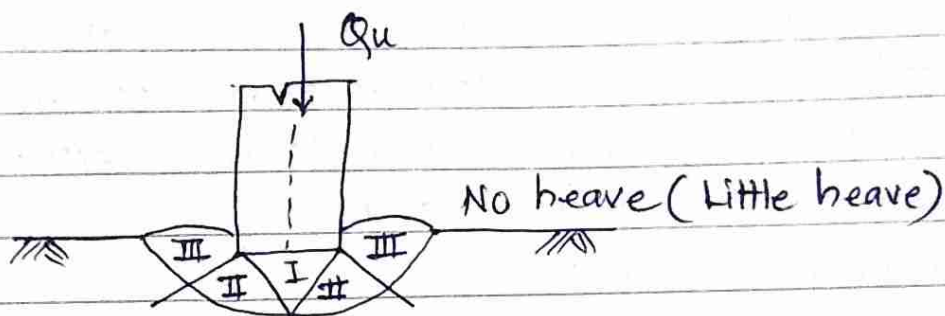
Modes of shear failure -

- (i) General shear failure
- (ii) Local shear failure
- (iii) Punching shear failure.

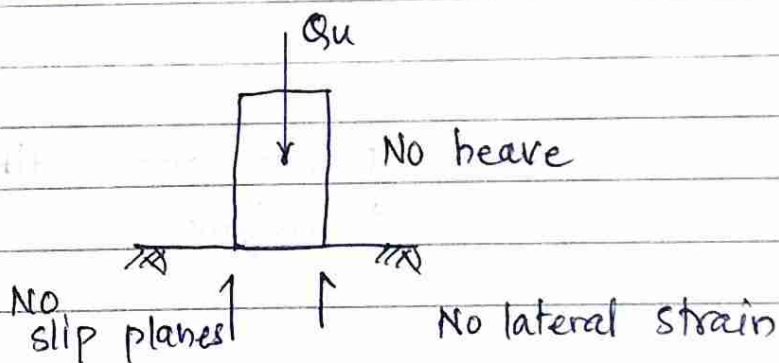
1) General shear failure



2) Local shear failure



3) Punching shear failure



Que- Write a short note on: comparison of shear failures

Sr. No.	General shear Failure	Local shear Failure	Punching shear Failure
01.	Dense sand, stiff and hard clays.	loose & medium sands, medium to soft clay.	very loose & soft saturated clays.

Sr. No.	General shear Failure	Local shear Failure	Punching shear failure
2.	$I_D > 65$, $N > 30$, $\phi > 36^\circ$	$I_D = 15-65$ $N = 10-30$, $\phi < 30^\circ$	$I_D < 15$, $N < 10$ $\phi < 10^\circ$
3.	Triaxial test shows pronounced peak and failure at small strain	No peak and failure at large strains	Failure with heavy compression.
4.	Base failure occurs at very small vertical strains accompanied by large lateral strains with surface heave.	It occurs at large vertical strain and very small lateral strains with no surface heave	No lateral strains, only large vertical strains.
5.	Sinking of base less than 7% base width	Sinking may range from 15-20% base width	Sinking may range from >20% base width.
6.	Failure zone is well defined	Failure zone is difficult to define	Failure is by sinking and indicates no slip planes.
7.	In load settlement curve, failure load is well defined and peak load is the ultimate load.	From load settlement graph, first indication of failure is large deformation in soil in early stage of loading which is difficult to distinguish during an incremental load test.	Load settlement curve indicates failure at very small loads and excessive settlements.

* 3. Methods of Determining Bearing capacity

Que: Explain the methods of determining Bearing capacity.
 → Following methods are available for the determination of bearing capacity of Foundation.

i) Bearing capacity tables in various building codes -
 Bearing capacity tables have been evolved by certain agencies and incorporated in building codes. They are mostly based on past experience and some investigations.

ii) Penetration Tests (SPT)

Penetration Tests are conducted with devices known as Penetrometer's which measure the resistance of soil to penetration, & it helps in determination of bearing capacity.

iii) Plate bearing Test (Plate load Test)

Plate bearing Test is load test conducted in the field on a plate, which is used for determining the bearing capacity of soil.

iv) Laboratory Tests :

These are simple, may be useful in arriving at bearing capacity, especially for pure clays.

v) Model & prototype tests -

These are very cumbersome and costly and are not usually practicable. Housel's approach is based on model tests.

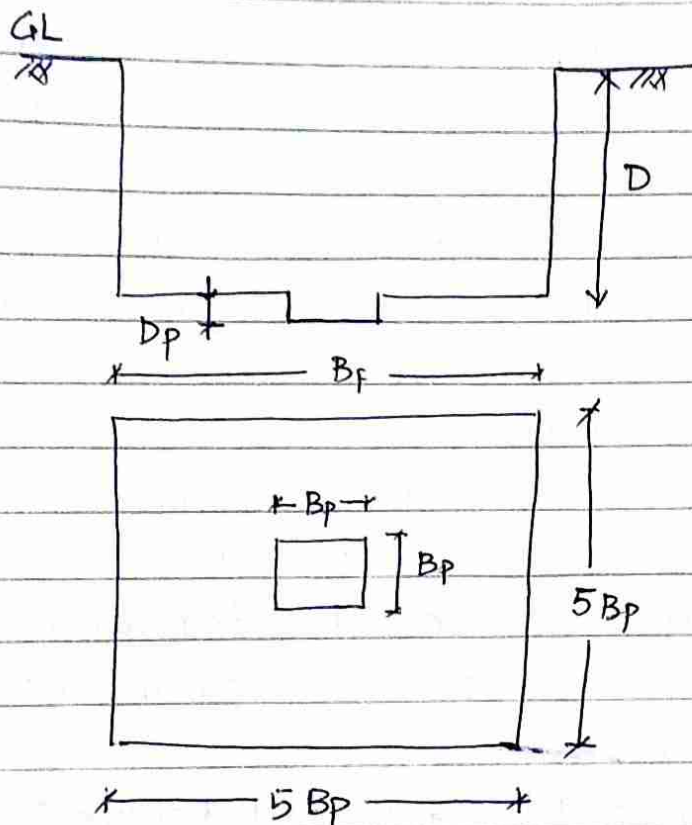
vi) Analytical methods -

A number of analytical approaches based on the work of Rankine, Fellenius, ~~Housel~~, Prandtl, Terzaghi, Meyerhof, skew skempton, Hansen & Bella may be used for determining bearing capacity of soil.

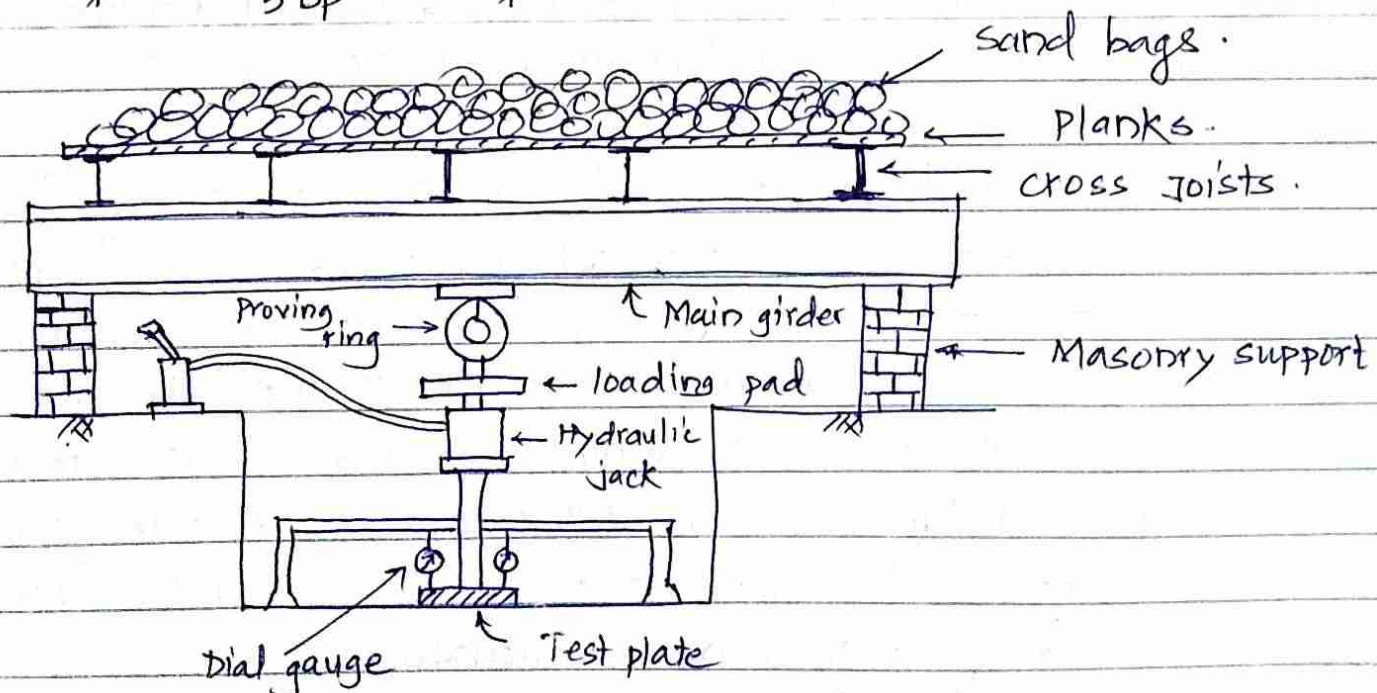
Plate load test (Field test for bearing capacity evaluation)

Write short note on plate load test.

Objective - To determine ultimate bearing capacity of soil and the probable settlement under given loading.



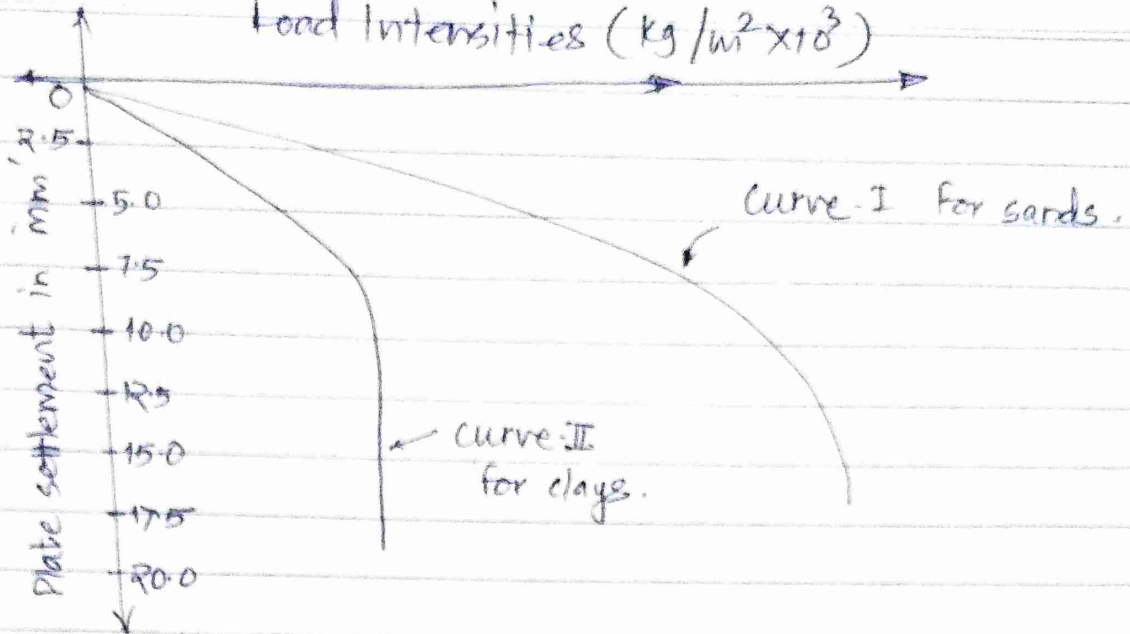
$$\frac{B_p}{B_f} = \frac{D_p}{D_f}$$



Procedure -

1. At first, pit is excavated of dimensions as shown in fig.
2. A thick plate square or circular plate of size 30 to 75 cm is placed at the centre below the bottom of test pit.
3. Hydraulic jack is placed over these plates and a proving ring is placed underneath steel truss and over hydraulic jacks.
4. By applying pressure through hydraulic jack, effort is made to lift the load on main steel girder. This causes compression of proving ring.
5. Compression of proving ring is proportional to the load applied. Compression of ring is noted by a dial gauges. Thus, indirectly the dial gauge will indicate the load applied.
6. To know the settlement of plate, three dial gauges are placed over the steel plates. The dial gauges themselves are supported by datum bar.
7. M.S. plate of 2.5 cm thickness is used for the test.
8. At the start, initial load of 70 gm/cm^2 is applied & released.
9. Then load in increments of 200 kg or $\frac{1}{5}^{\text{th}}$ of approx. load bearing capacity, whichever is less is applied, till about 1.5 times the expected ultimate bearing capacity.
10. Each load is maintained constant until the settlement rate is less than 0.002 mm/hr . Settlement are recorded at following time interval 1, 2, 5, 10, 20, 40, 60 min, 2 hr 3 hr 24 hr.
11. The load - settlement graph is plotted from which ultimate bearing capacity & hence allowable bearing capacity can be determined.

Load Intensities ($\text{kg}/\text{m}^2 \times 10^3$)



- Ultimate load can be determined as follows;

select settlement = $\frac{1}{5}^{\text{th}}$ of width of plate & corresponding load to this settlement on the graph is taken as ultimate load.

- Settlement of footing on a cohesionless soil can be determined from test plate at same load intensity by the following equation;

$$\frac{S_f}{S_p} = \left[\frac{B_f (B_p + 0.3)}{B_p (B_f + 0.3)} \right]^2$$

Where;

S_f - settlement in foundation, S_p - settlement in plate

B_f = width of Foundation, B_p = width of plate

- For clayey soil;

$$\frac{S_f}{S_p} = \frac{B_f}{B_p}$$

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Que Enlist the limitations of plate Bearing Test.

- 1. Test is carried on a small area & the results are used for larger area.
2. A plate load test is of short duration. The settlement measured is only immediate settlement. In granular soil immediate settlement can be taken as the total settlement while in cohesive soils total settlement including consolidation cannot be predicted through this test.
3. If q is the intensity of loading on plate, then at depth equal to twice the width of plate ($2 B_p$) stress reduces to $0.1 q$ which is negligible. Thus with this test, soil up to a depth of twice the width of steel plate is tested. But in practice, if the width of footing is far more than B_p . Hence if loose weak soil present within a depth of $2 B_f$ then results obtained from the plate bearing test will not hold good.
4. It is difficult to carry out test at greater depth.
5. The results of the test carried on small square plate cannot be used for continuous footing.
6. No consideration of rise in water table is there in plate load test.

* 4. Bearing capacity analysis -

Que- Enlist the assumptions made in Terzaghi's theory for bearing capacity determination and state Terzaghi's equation for bearing capacity with meaning of each term.

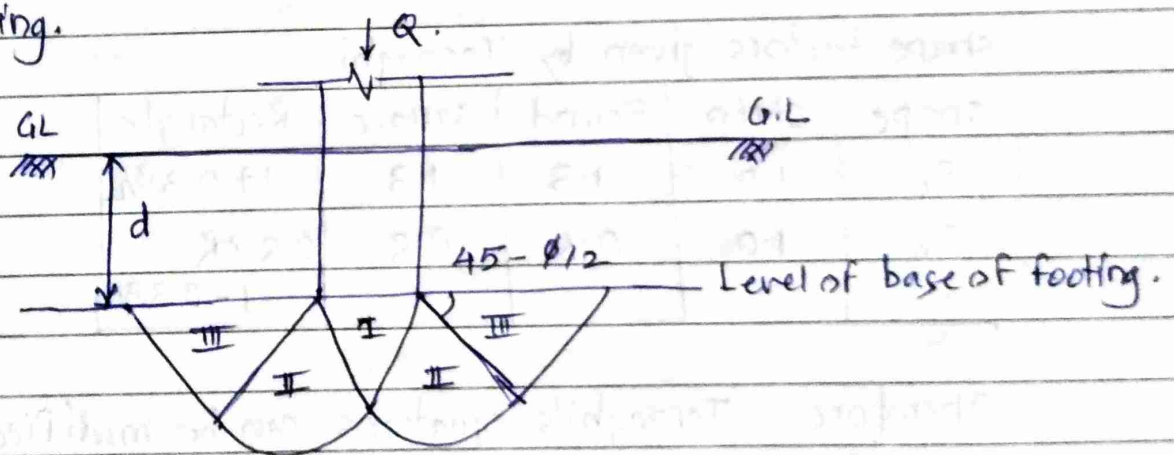
Que- State Terzaghi's bearing capacity equation for strip, circular, square footing and describe all the terms.

→

* Terzaghi's theory -

• Assumptions -

1. Base of footing is rough.
2. Footing is laid at shallow depth.
3. Load on footing is vertical and is uniformly distributed
4. Footing is strip ($L \gg B$)
5. Shear strength of soil is governed by $s = c + \sigma \tan \phi$
6. Soil is homogeneous, isotropic and incompressible.
7. There is no friction between soil and footing.
8. Rupture surface extend only upto the level of base of footing.



• Theory -

Bearing capacity depends on shearing resistance along the boundary of failure zone which is divided into three parts,

- (1) cohesion
- (2) Frictional resistance resulting from surcharge at footing level.
- (3) Frictional resistance resulting from self wt. of wedge of soil.

- Terzaghi has derived an equation for bearing capacity considering equilibrium of wedge under footing,

$$q_u = CN_c + \gamma_1 DN_q + 0.5 \gamma_2 BN_\gamma$$

Here;

q_u = ultimate bearing capacity of soil

γ_1 = Unit wt. of soil above footing

γ_2 = Unit wt. of soil below footing

d = Depth of footing.

B = Width of footing.

N_c, N_q, N_γ = Terzaghi's Bearing capacity factors.

Above equation is for strip footing and for general shear failure.

- Specialization of Terzaghi's equations (consideration of shape of footing)

shape factors given by Terzaghi

shape	strip	Round	square	Rectangle
S_c	1.0	1.3	1.3	$1 + 0.3B/L$
S_γ	1.0	0.6	0.8	0.8 OR $1 - 0.3B/L$

Therefore; Terzaghi's equations can be modified as below,

- ① For strip footing

$$q_u = CN_c + \gamma_1 DN_q + 0.5 \gamma_2 BN_\gamma$$

- ② For round (circular) footing

$$q_u = 1.3 CN_c + \gamma_1 DN_q + 0.3 \gamma_2 BN_\gamma$$

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③ For square footing

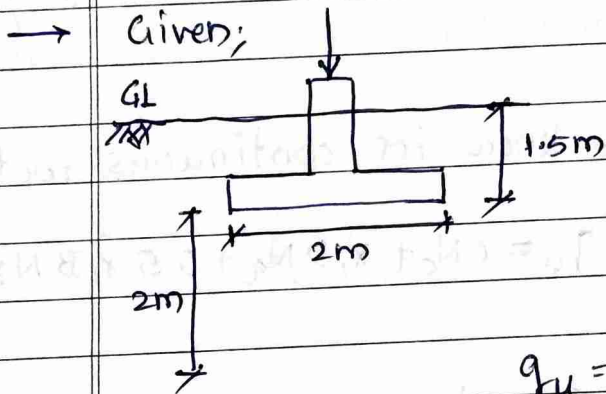
$$q_u = 1.3 C N_c + \gamma_1 D N_q + 0.4 \gamma_2 B N_\gamma$$

④ For rectangular footing

$$q_u = (1 + 0.3 B/L) C N_c + \gamma_1 D N_q + (1 - 0.3 B/L) 0.5 \gamma_2 B N_\gamma$$

* Numericals -

1. A rectangular footing $2\text{m} \times 3\text{m}$ rests on c- ϕ soil with its base at 1.5m below the ground surface, calculate
 (1) Net ultimate bearing capacity (2) SBC For
 $N_c = 37.2$, $N_q = 22.5$, $N_\gamma = 19.7$, $F_s = 3$, $\gamma = 18\text{ KN/m}^3$,
 $c = 10\text{ KN/m}^2$, $L = 3$, $D = 1.5\text{m}$, $B = 2\text{m}$.



We know;

For rectangular footing;

$$q_u = (1 + 0.3 B/L) C N_c + \gamma_1 D N_q + (1 - 0.3 B/L) 0.5 \gamma_2 B N_\gamma$$

$$\begin{aligned} q_u &= (1 + 0.3 \times 2/3) \times 10 \times 37.2 + 18 \times 1.5 \times 22.5 \\ &\quad + (1 - 0.3 \times 2/3) \times 0.5 \times 18 \times 2 \times 19.7 \\ &= (1.2 \times 10 \times 37.2) + (18 \times 1.5 \times 22.5) + (0.8 \times 0.5 \times 18 \times 2 \times 19.7) \\ &= 1.33 \times 10^3 \text{ KN/m}^2 \end{aligned}$$

Net ultimate bearing capacity (q_{nu})

$$\begin{aligned} q_{nu} &= q_u - \gamma_1 D \\ &= (1.33 \times 10^3) - (18 \times 1.5) \\ &= 1.31 \times 10^3 \text{ KN/m}^2 \end{aligned}$$

Net safe bearing capacity (q_{ns})

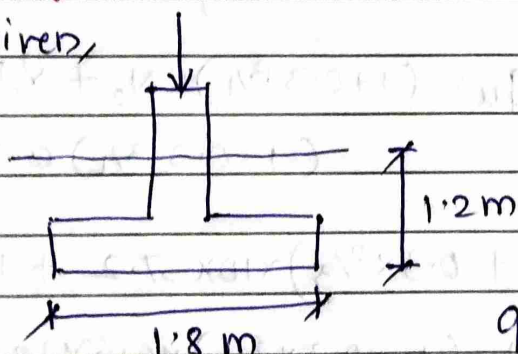
$$q_{ns} = \frac{q_{nu}}{F} = \frac{1.31 \times 10^3}{3} = 436.67 \text{ KN/m}^2$$

safe bearing capacity (q_s)

$$\begin{aligned} q_s &= q_{ns} + \gamma_1 D \\ &= 436.67 + (18 \times 1.5) \\ &= 463.67 \text{ KN/m}^2 \end{aligned}$$

2. Compute safe bearing capacity of a continuous footing 1.8 m wide and located at depth of 1.2 m below ground level in soil having unit weight $\gamma = 20 \text{ KN/m}^3$, $C = 20 \text{ KN/m}^2$ and $\phi = 20^\circ$. Assume factor of safety 2.5. Terzaghi's bearing capacity factors For $N_c = 17.7$, $N_q = 7.4$, $N_\gamma = 5.0$

→ Given,



We know for continuous footing

$$q_u = C N_c + \gamma_1 D N_q + 0.5 \gamma_1 B N_\gamma$$

$$\begin{aligned} q_u &= (20 \times 17.7) + (20 \times 1.2 \times 7.4) + \\ &\quad (0.5 \times 20 \times 1.8 \times 5) \end{aligned}$$

$$C = 20 \text{ KN/m}^2$$

$$\phi = 20^\circ$$

$$\gamma = 20 \text{ KN/m}^3$$

$$FS = 2.5$$

$$N_c = 17.7, N_q = 7.4$$

$$N_\gamma = 5.0$$

Net ultimate bearing capacity

$$q_{nu} = q_u - \gamma_1 D$$

$$= 621.6 - (20 \times 1.2)$$

$$= 597.6 \text{ KN/m}^2$$

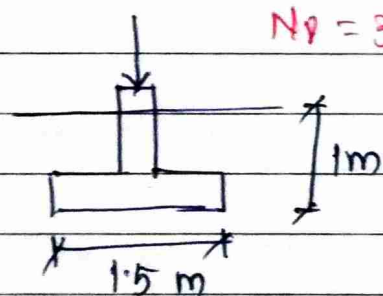
Net safe bearing capacity

$$q_{ns} = \frac{q_{nu}}{F_s} = \frac{587.6}{2.5} = 239.04 \text{ KN/m}^2$$

safe bearing capacity

$$\begin{aligned} q_s &= q_{ns} + \gamma_1 D \\ &= 239.04 + (20 \times 1.2) \\ &= 263.04 \text{ KN/m}^2 \end{aligned}$$

3. A square foundation is $1.5 \text{ m} \times 1.5 \text{ m}$ in plan. The soil supporting the foundation has friction angle $\phi = 20^\circ$ and $c = 52 \text{ KN/m}^2$. The unit wt. of soil $\gamma = 17.8 \text{ KN/m}^3$. Determine the allowable gross pressure on foundation with $FOS = 4$. Assume that the depth of foundation is 1 m and general shear failure occurs in soil. Take $N_c = 17.69$, $N_q = 7.44$, $N_\gamma = 3.64$.



$$N_\gamma = 3.64$$

Given;

$$B = 1.5 \text{ m}, \phi = 20^\circ, c = 52 \text{ KN/m}^2$$

$$\gamma = 17.8 \text{ KN/m}^3, FOS = 4, D = 1 \text{ m}$$

Allowable gross pressure means safe pressure

We know;

For square Footing -

$$\begin{aligned} q_u &= 1.3 c N_c + \gamma_1 D N_q + 0.4 \gamma_2 B N_\gamma \\ &= (1.3 \times 52 \times 17.69) + (17.8 \times 1 \times 7.44) + (0.4 \times 17.8 \times 1.5 \times 3.64) \\ &= 1367.15 \text{ KN/m}^2 \end{aligned}$$

$$q_{nu} = q_u - \gamma_1 D = (1367.15 - 17.8 \times 1) = 1349.35 \text{ KN/m}^2$$

$$q_{ns} = \frac{q_{ny}}{FS} = \frac{1349.35}{4} = 337.34 \text{ kN/m}^2$$

$$\begin{aligned} q_s &= q_{ns} + \gamma_1 D \\ &= 337.34 + (17.8 \times 1) \\ &= 355.14 \text{ kN/m}^2 \end{aligned}$$

4. A circular Footing is resting on a stiff saturated clay with $q_u = 250 \text{ kN/m}^2$. The depth of foundation is 2m. Determine the dia. of the footing if the column load is 600 kN. Assume FS as 2.5. The bulk unit wt. of soil is 20 kN/m^3 . Use Terzaghi's Formula.

→ Given,

Circular Footing, $q_u = 250 \text{ kN/m}^2$, $D = 2\text{m}$, dia. = ? $Q = 600 \text{ kN}$,
 $FS = 2.5$, $\gamma_b = 20 \text{ kN/m}^3$, For cohesive soil $N_c = 5.7$,
 $N_q = 1$, $N_\gamma = 0$

We know;

$$c = q_u/2 = 250/2 = 125 \text{ kN/m}^2$$

$$q_u = 1.3 C N_c + \gamma_1 D N_q + 0.3 \gamma_b N_\gamma$$

$$\begin{aligned} &= (1.3 \times 125 \times 5.7) + (20 \times 2 \times 1) + (0.3 \times 20 \times \text{dia.} \times 0) \\ &= 966.25 \text{ kN/m}^2 \end{aligned}$$

We know; $q_u = \frac{Q}{\text{Area}}$

$$\therefore \text{Area} = \frac{600}{966.25} = \frac{\pi}{4} \times d^2 = \frac{600}{966.25}$$

$$\therefore d = \sqrt{\frac{600 \times 4}{\pi \times 966.25}} = \underline{\underline{0.89 \text{ m}}}$$

Mayerhof's Analysis -

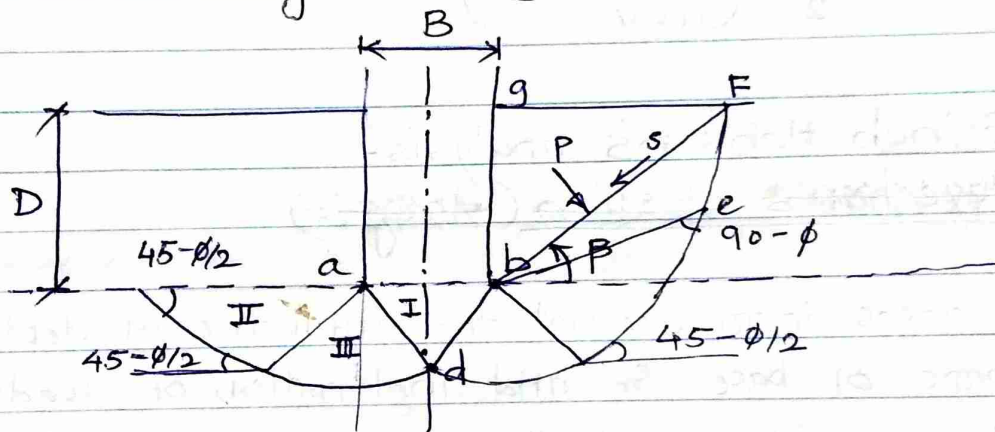
Ques- Write short note on Mayerhof's theory.

→ Mayerhof considered shape factor, Inclination factor and depth factor during finding out bearing capacity.

$$q_{ru} = c N_c s_c i_c d_c + \gamma N_q s_q i_q d_q + 0.5 \gamma B N_\gamma s_\gamma i_\gamma d_\gamma$$

Here: s_c, s_q and $s_\gamma \Rightarrow$ shape factors
 i_c, i_q and $i_\gamma \Rightarrow$ Inclination factors.
 d_c, d_q and $d_\gamma \Rightarrow$ Depth factors.

Mayerhof's Analysis & Terzaghi's Analysis comparison.



Terzaghi's theory $\longleftrightarrow$ Mayerhof theory

- In Mayerhof's analysis, abd is the elastic zone, bde is the radial shear zone and $befg$ is the zone of mixed shear.
- To simplify the analysis, Mayerhof introduced a parameter β , the angle to define line bf joining point b to f .
- The resultant effects of wedge of soil bgf is represented by the normal and tangential stresses p and s on bf .
- Plane bf is termed as the equivalent free surface and p and s are termed as the equivalent free surface stresses.

- Angle β' increases with depth, and becomes 90° for deep foundation.
- Even Terzaghi's analysis differs from Meyerhofs analysis. Terzaghi's general equation can be used with Meyerhofs bearing capacity Factors.

Terzaghi's	Meyerhofs
1. It considers lateral stress coefficients in passive state	1. It involves the lateral stress coefficient which ranges b/n the active & passive values, depending on unit weight, angle of friction and deformation condition.
2. Shear zone does not extend above base level.	2. The shear zone extends above the base level.
3. Suitable for shallow Foundation only	3. He extended the analysis of plastic equilibrium of a surface footing to a shallow and deep foundation.

* IS (6403-1981) Method of Finding bearing Capacity. For c- ϕ soil.

- According to IS (6403-1981) method, ultimate net bearing capacity of strip footing is given by following equation:

$$q_{na} = C N_c + q (N_q - 1) + 0.5 \gamma B N_\gamma$$

where; q = Effective surcharge at the base level of foundation.

OR

$$q_{nu} = C N_c S_c i'_c d_c + q (N_q - 1) S_q i'_q d_q + 0.5 \gamma B N_\gamma S_\gamma i'_\gamma d_\gamma$$

Skempton's Analysis

Que Explain Skempton's analysis of determination of bearing capacity.

→

- $q_{nu} = C N_c$

Value of N_c according to Skempton,

Case-i) when depth (D)=0

$$N_c = 5.14 \leftarrow \text{for strip footing}$$

$$N_c = 2.60 \leftarrow \text{for square footing}$$

$$N_c = 5.14 (1 + 0.2 B/L) \leftarrow \text{Rectangular footing.}$$

Case-ii) when, $D/B < 2.5$

$$N_c = [1 + 0.2 D/B] \times (N_c)_{\text{surface}}$$

Case-iii) when, $D/B > 2.5$

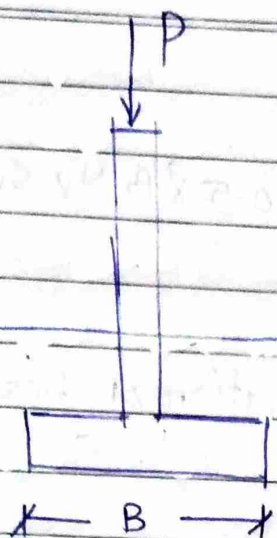
$$N_c = 1.5 \times (N_c)_{\text{surface}}$$

5 *

Effect of water table and eccentricity on Bearing capacity.

Effect of water table on B.C.

- Bearing capacity of soil reduced.
- To calculate the bearing capacity and the effect of water table on it, we have to apply water table reduction factors.



Approximate method -

For any intermediate position of water table,

$$q_u = cN_c + \gamma D N_q R_{w1} + 0.5 \gamma B N_q R_{w2}$$

$$R_{w1} = 0.5 \left[1 + \frac{d_1}{D} \right], \quad R_{w2} = 0.5 \left[1 + \frac{d_2}{B} \right]$$

Here, d_1 = Distance of water level measured from ground level

d_2 = Distance of water level measured from base of footing.

Note: d_1 should not be more than D i.e. max. value of $d_1 = D$, therefore max. value of $R_{w1} = 1$

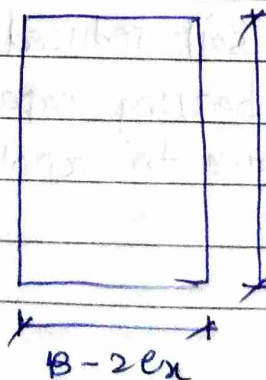
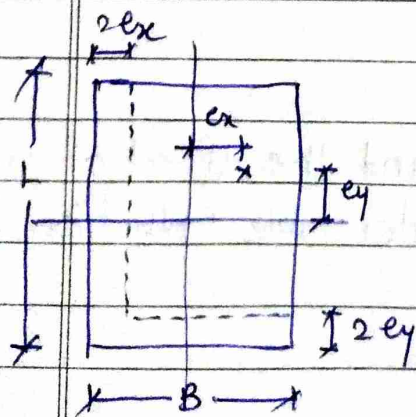
In the same manner

d_2 should not be more than B i.e. max. value of $d_2 = B$, therefore, max. value of $R_{w2} = 1$

Effect of eccentricity on bearing capacity.

- If total load on footing acts eccentrically, the width B' and length L' should be reduced as under

$$B' = B - 2e_x, \quad L' = L - 2e_y$$



New area

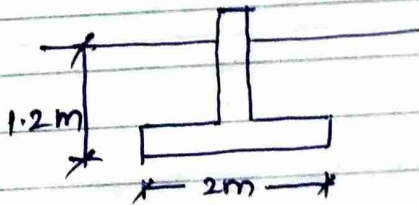
$$= B' \times L'$$

$$Q = 280 \text{ kN}$$

Numerical examples based on,
Effect of water table & eccentricity on Bearing capacity

10. compare the effect of W.T. on B.C. of soil for footing 2m wide, located at a depth of 1.2m in sand with following data
- i) Water table is 4m below GL
 - ii) W.T. is 1.2m below GL
- $\gamma = 16.8 \text{ kN/m}^3$ $\gamma_{sat} = 19.5 \text{ kN/m}^3$
 $N_q = 41.4$ $N_{\gamma} = 42.4$

→ Given,



case i) When W.T. @ 4m below GL

No effect of w.t. on bearing capacity of soil, as $R_{w1} = 1$
 & $R_{w2} = 1$

$$q_u = C N_c + \gamma \cdot D N_q R_{w1} + 0.5 \gamma B N_{\gamma} R_{w2}$$

$$= 0 + (16.8 \times 1.2 \times 41.4 \times 1) + (0.5 \times 16.8 \times 2 \times 42.4 \times 1)$$

$$q_u = 1546.94 \text{ kN/m}^2$$

case ii) When W.T. @ 1.2m below GL

$$R_{w1} = 0.5 \left[1 + \frac{d_1}{D} \right]$$

$$= 0.5 \left[1 + \frac{1.2}{1.2} \right]$$

$$= 1$$

$$R_{w2} = 0.5 \left[1 + \frac{d_2}{B} \right]$$

$$= 0.5 \left[1 + 0 \right]$$

$$= 0.5$$

$$\therefore q_u = cN_c + \gamma \cdot D N_q + 0.5 \gamma_{\text{sat}} B N_{\gamma} R_{w2}$$

$$= 0 + (16.8 \times 1.2 \times 41.4 \times 1) + (0.5 \times 19.5 \times 2 \times 42.4 \times 0.5)$$

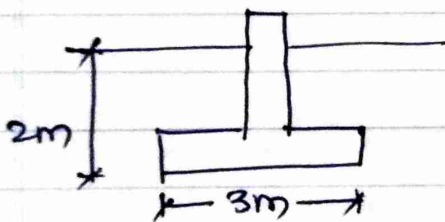
$$q_u = 1248 \text{ KN/m}^2$$

11. A 3.0 m square footing is located in a dense sand at a depth of 2 m. Determine the ultimate bearing capacity for the following water table position.

- ① At ground surface
- ② At footing level
- ③ At 1 m below the footing.

The moist unit wt. of sand above the w.T. is 18 KN/m^3 and saturated unit wt. is 20 KN/m^3 , $\phi = 35^\circ$, $c = 0$, $N_q = 33$, $N_{\gamma} = 34$

→ Given



case-i) At ground surface,

$$R_{w1} = 0.5 \left[1 + \frac{d_1}{D} \right] \quad R_{w2} = 0.5 [1 + 0]$$

$$= 0.5 \left[1 + \frac{0}{2} \right] \quad = 0.5$$

$$= 0.5$$

$$\therefore q_u = 1.3 cN_c + \gamma_{\text{sat}} D N_q R_{w1} + 0.4 \gamma_{\text{sat}} B N_{\gamma} R_{w2}$$

$$= (1.3 \times 0) + (20 \times 2 \times 33 \times 0.5) + (0.4 \times 20 \times 3 \times 34 \times 0.5)$$

$$= 1068 \text{ KN/m}^2$$

case - ii) At footing level

$$R_{w1} = 0.5 \left[1 + \frac{d_1}{D} \right] \quad R_{w2} = 0.5 [1 + 0]$$

$$= 0.5 \left[1 + \frac{0}{2} \right] = 1 \quad = 0.5$$

$$\therefore q_u = 1.3 C N_c + \gamma D N_q R W_1 + 0.4 \gamma_{sat} B N_\gamma R W_2$$

$$= 0 + (18 \times 2 \times 33 \times 1) + (0.4 \times 20 \times 3 \times 34 \times 0.5)$$

$$q_u = 1596 \text{ KN/m}^2$$

Case-iii) When W.T. @ 1m below the footing,

$$R W_1 = 1 \quad R W_2 = 0.5 \left[1 + \frac{1}{3} \right]$$

$$= 0.67$$

$$q_u = 1.3 C N_c + \gamma D N_q R W_1 + 0.4 \gamma_{avg} B N_\gamma R W_2$$

$$= 0 +$$

$$\gamma_{avg} \times 3 = (9 \times 1) + (\gamma_{sat} \times 2)$$

$$\gamma_{avg} = \frac{(18 \times 1) + (20 \times 2)}{3} = 19.33 \text{ KN/m}^3$$

$$\therefore q_u = 0 + (18 \times 2 \times 33 \times 1) + (0.4 \times 19.33 \times 3 \times 34 \times 0.67)$$

$$= 1716.40 \text{ KN/m}^2$$

12. Compute the ultimate load that an eccentrically loaded square footing of width 2.1 m with an eccentricity of 0.35 m can carry at a depth of 0.5 m in a soil with $c = 9 \text{ KN/m}^2$, $\gamma = 20 \text{ KN/m}^3$ and $\phi = 36^\circ$, $N_c = 52$, $N_q = 35$ and $N_\gamma = 42$.

$$\rightarrow B = 2.1 \text{ m}, c = 9 \text{ KN/m}^2, e = 0.35$$

$$B' = B - 2e$$

$$= 2.1 - (2 \times 0.35)$$

$$= 1.4$$

$$q_u = 1.3 c N_c + \gamma D N_q + 0.4 \gamma B' N_\gamma$$

$$= (1.3 \times 9 \times 52) + (20 \times 0.5 \times 35) + (0.4 \times 20 \times 1.4 \times 42)$$

$$= 1428.8 \text{ KN/m}^2$$

Ultimate load, $Q = q_f \times A'$

$$= 1428.8 \times 1.4 \times 1.4$$

$$Q = 2800 \text{ KN.}$$

*6. Settlement Analysis

Causes of settlement in soil below foundation

- Lowering of water table and dewatering.
- Static load on foundation can cause consolidation settlement.
- Dynamic loading can cause shocks and vibrations e.g. Earthquake.
- Deep excavation, pile driving, mining activities (construction activities) are responsible for the settlement of soil below foundation.

Types of settlement

There are various types of settlement

1. Immediate settlement / Initial or elastic settlement

- Takes place immediately after applying load.
- Occurs in all type of soil.
- Depends on elastic properties of soil, rigidity, size and shape of foundation.
- Given by

$$s_i = B I q \left(\frac{1 - u^2}{E} \right)$$

Here, q = Contact pressure @ base of footing.

B = width of footing

I = Influence factor

E = Modulus of elasticity of soil.

u = Poisson's ratio for the soil.

Examples -

1. Determine immediate settlement of square footing of 1m size founded at depth of 1m in a soil with $E = 10000 \text{ kN/m}^2$, $\mu = 0.3$ the footing is subjected to pressure of 200 kN/m^2 . Assume influence coefficient = 1.

→ Given data,

$$B = 1 \text{ m}$$

We know;

$$E = 10000 \text{ kN/m}^2$$

$$\mu = 0.3$$

$$q = 200 \text{ kN/m}^2$$

$$I = 1$$

$$S_i = B I q \left(\frac{1 - \mu^2}{E} \right)$$

$$= 1 \times 1 \times 200 \left(\frac{1 - 0.3^2}{10000} \right)$$

$$S_i = 0.0182 \text{ m}$$

2. A square footing on sand at 2m depth shows an elastic settlement of 5.5 mm. Under a loading of 200 kN/m^2 . How much a footing would be settle if has to carry load of 150 kN/m^2 .

→ Given data,

$$q_1 = 200 \text{ kN/m}^2, S_{i1} = 5.5 \text{ mm}, q_2 = 150 \text{ kN/m}^2, S_{i2} = ?$$

$$S_i = B I q \left(\frac{1 - \mu^2}{E} \right)$$

As B, I, μ, E are constant

we can write

$$S_i \propto q$$

$$\frac{S_{i1}}{q_1} = \frac{S_{i2}}{q_2} \Rightarrow \frac{5.5}{200} = \frac{S_{i2}}{150}$$

~~5/2~~

$$S_{i2} = 4.125 \text{ mm}$$

2) Consolidation settlement -

i) Primary consolidation settlement

- settlement which takes place due to expulsion of water from the soil.
- It is applicable for saturated clay.
- Given by

$$S_c = \frac{C_c \cdot H}{1+e_0} \log_{10} \frac{P+\Delta P}{P}$$

$$= \frac{a_v}{1+e_0} \times \Delta P \times H$$

$$= m_v \times \Delta P \times H$$

Here; S_c = Primary consolidation settlement

H = Thickness of compressive stratum

a_v = Coeff. of compressibility.

e_0 = Initial void ratio

C_c = Compression Index

m_v = coeff. of vol^m compressibility

P = Initial pressure at centre of layer

ΔP = Increase in pressure at centre of layer due to load on footing.

Example: 1. The overburden pressure at the middle of 7.5 cm thick clayey layer increased from 2 kg/cm² to 3.5 kg/cm². Find settlement due to pressure consolidation assuming liquid limit and initial void ratio of clay as 36% and 0.82 resp.

→ Given,

$$H = 7.5 \text{ cm}$$

$$w_L = 36\% = 0.36$$

$$P = 2 \text{ kg/cm}^2$$

$$\Delta P = 1.5 \text{ kg/cm}^2$$

$$e_0 = 0.82$$

We have;

$$S_c = C_c \times \frac{H}{1+e_0} \log_{10} \left(\frac{P+\Delta P}{\Delta P} \right)$$

If C_c is not given; then use following formula -

$$C_c = 0.009 (w_L - 10) \quad \text{--- for clay}$$

$$= 0.007 (w_L - 10) \quad \text{--- for sand}$$

$$C_c = 0.009 (36 - 10)$$

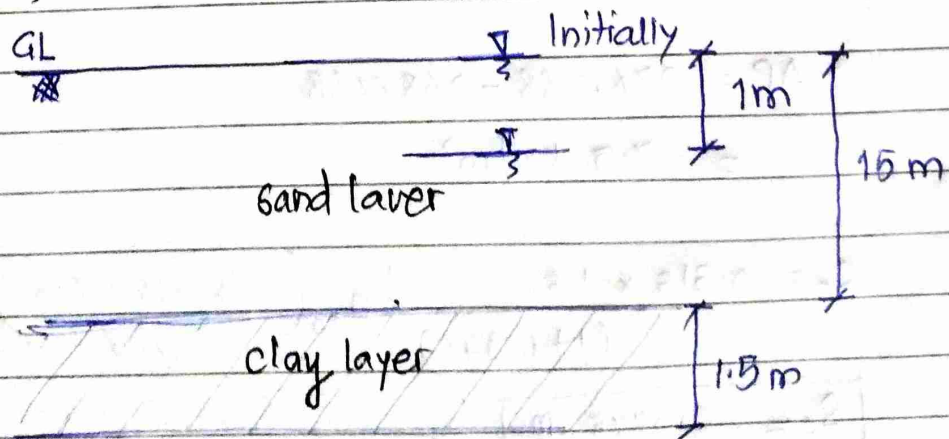
$$= 0.234$$

$$S_c = 0.234 \times \frac{7.5}{1+0.82} \log_{10} \left(\frac{2+1.5}{1.5} \right)$$

$$S_c = 0.3548 \text{ cm}$$

- 2) Determine the consolidation settlement of clay layer 1.5 m thick at depth of 15 m below G.L. If water table is lowered from ground level to 1 m depth. The clay layer underlies the sand bed. The soil properties are as under, $\gamma_{sat} = 17.5 \text{ kN/m}^3$, $\gamma_b = 10 \text{ kN/m}^3$, $S_g = 2.65$, w for clay = 35%, w_L of clay = 45%, γ_d of clay = 9 kN/m^3 , specific gravity for clay = 2.7

Given data,



$$S_c = C_c \cdot \frac{H}{1+e_0} \log_{10} \left(\frac{P+\Delta P}{\Delta P} \right)$$

$$C_c = 0.009 (\omega_L - 10)$$

$$= 0.009 (45 - 10)$$

$$= 0.315$$

$$s \times e = \omega \times G$$

$$e = \frac{\omega \times G}{s} = \frac{0.35 \times 2.7}{1} = 0.945$$

$$\gamma_{sat} = \left(\frac{G+e}{1+e} \right) \gamma_w$$

$$= \left(\frac{2.7+0.95}{1+0.95} \right) \times 9.81 = 18.384 \text{ KN/m}^3$$

$$P_1 = [\gamma_{sat(sand)} \times D_{(sand)}] + [\gamma_{sat(clay)} \times D_{(clay)}]$$

$$= (15 \times 17.5) + (18.384 \times 0.75)$$

$$P_1 = 276.288 \text{ KN/m}^2$$

$$\text{Effective stress} = (1 \times 10) + (14 \times 17.5) + (0.75 \times 18.384)$$

$$= 268.788 \text{ KN/m}^2$$

$$\Delta P = P_1 - P_2$$

$$\Delta P = 276.288 - 268.788$$

$$= 7.5 \text{ KN/m}^2$$

$$S_c = 0.315 \times 1.5 \log_{10} \left(\frac{276.288 - 7.5}{7.5} \right)$$

$$S_c = 0.3775 \text{ m}$$

ii) secondary consolidation settlement -

- Occurs at very slow & negligible rate after completion of initial consolidation.
- Also known as creep settlement.
- common in every sensitive clay.
- Given by,

$$S_s = C_\alpha \cdot H \log_{10} \left(\frac{t_p + \Delta t}{t_p} \right)$$

Here;

C_α = Coefficient of secondary consolidation.

H = Thickness of clay layer.

Δt = Time increment producing ' S_s '.

t_p = Time taken for completing primary consolidation.

3) Uniform settlement

Every part of structure settles by same amount.

4) Non-uniform settlement -

- Diff. parts of structure settles by unequal amount.
- Diff. between settlement at any two points is known as differential settlement.
- Differential settlement per unit length = Tilt OR Angular distortion
- Causes -

1) Large loaded areas on flexible foundation.

2) Non-uniform soil.

3) Non-uniform loading.

5) Tolerable settlement (Allowable settlement)

- The settlement which does not causes any effect on performance of structure.